

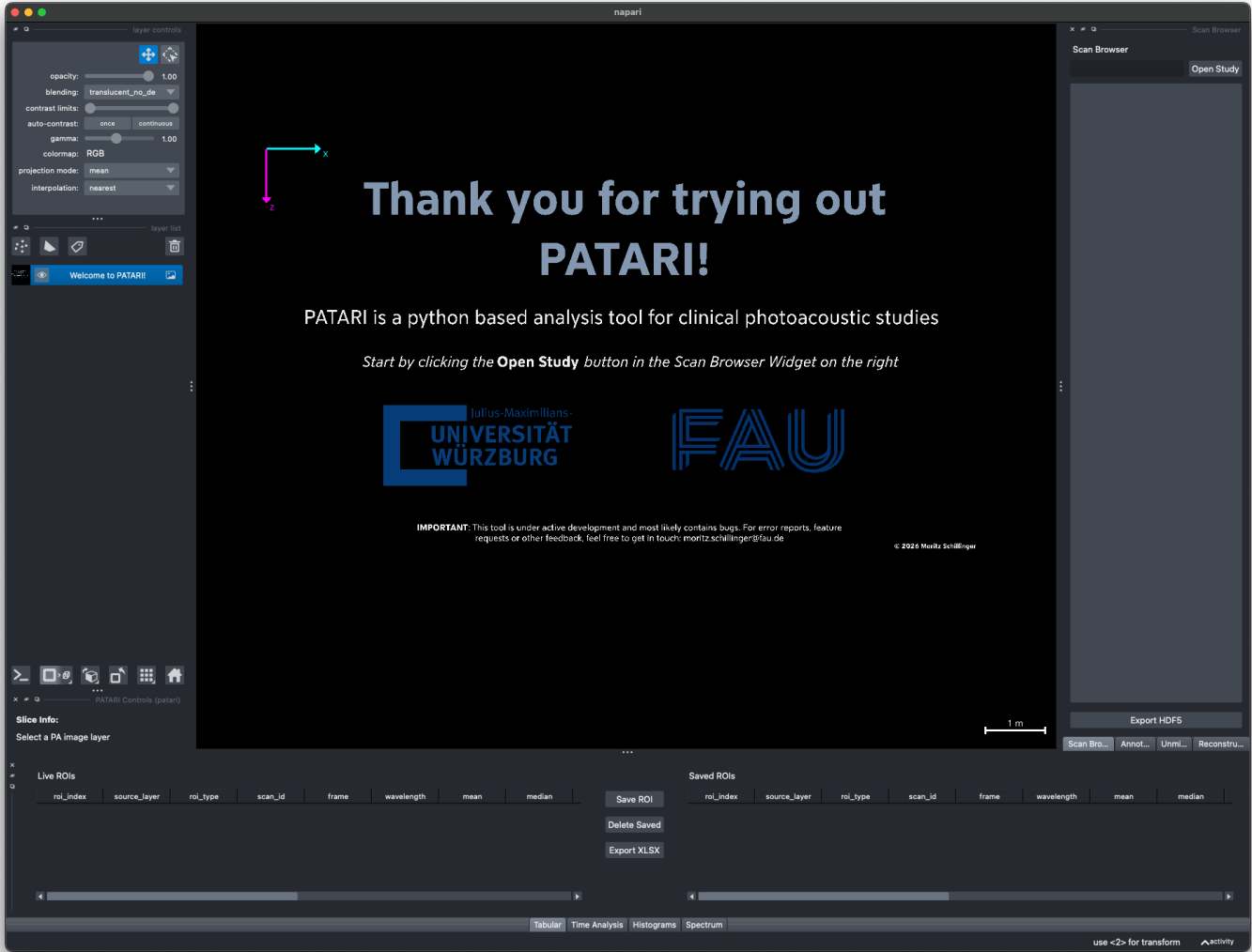
PATARI v0.1 Overview

Analysis Tool for Clinical Photoacoustic Imaging

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- An open source GUI tool that covers the most common analysis workflows for clinical photoacoustic studies
 - Target group: medical doctors / non-technical users
 - Pure-python based
 - Runs on Windows / MacOS / Linux
 - No additional dependencies, easy installation
 - Easily extendable and configurable
 - Relies on established frameworks:
 - NAPARI (GUI)
 - PATATO (Data I/O, Processing)
 - Modular architecture:
 - Can be adapted to individual needs
 - As simple as possible, as complex as necessary

Implemented or planned

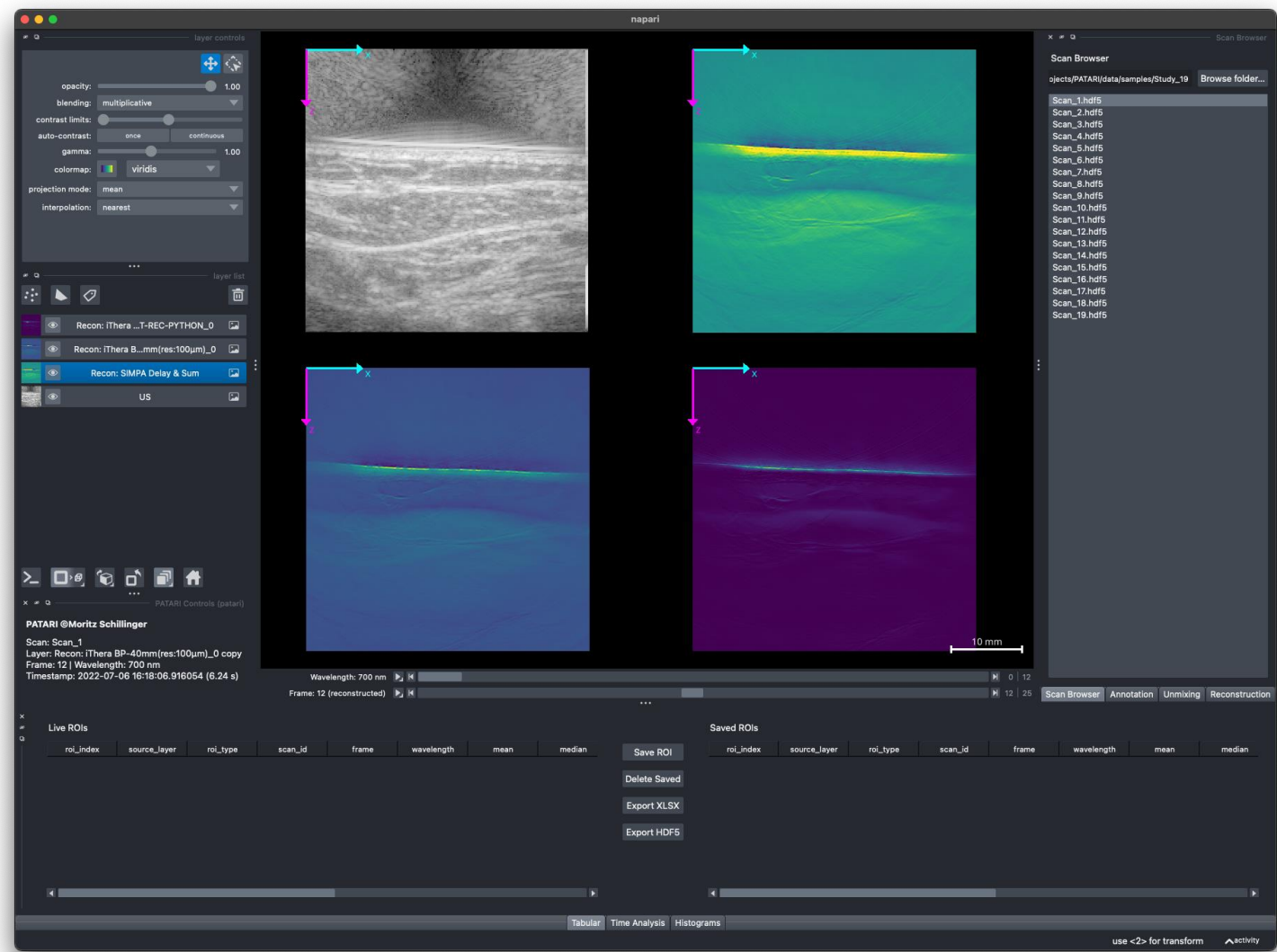
- Supports different file formats + easy extension to new ones
 - Native iThera (including iLabs annotations)
 - PATATO
 - IPASC
 - SIMPA
- Seamless scrolling through frames and wavelengths
- Overlay / side by side view of US and PA images
- Overlay / side by side view of different reconstructions
- Postprocessing / reconstruction / unmixing (provided by PATATO)
- Export to HDF5 / MS Excel
- **Quantitative image analysis:**
 - ROI placement and intensity extraction using NAPARI shapes
 - Import / export of ROIs + customizable default ROIs
 - Auto-updating live ROI table / Persistent saved ROI table
 - Plot PA intensity over time for multiple ROIs
 - Histogram view for multiple ROIs
- **Automation:**
 - Deep learning based tissue segmentation from US image (<https://doi.org/10.1117/12.3049067>)
 - Automatic ROI placement in target class (i.e. muscle, fat layer, ...)
 - Batch processing mode



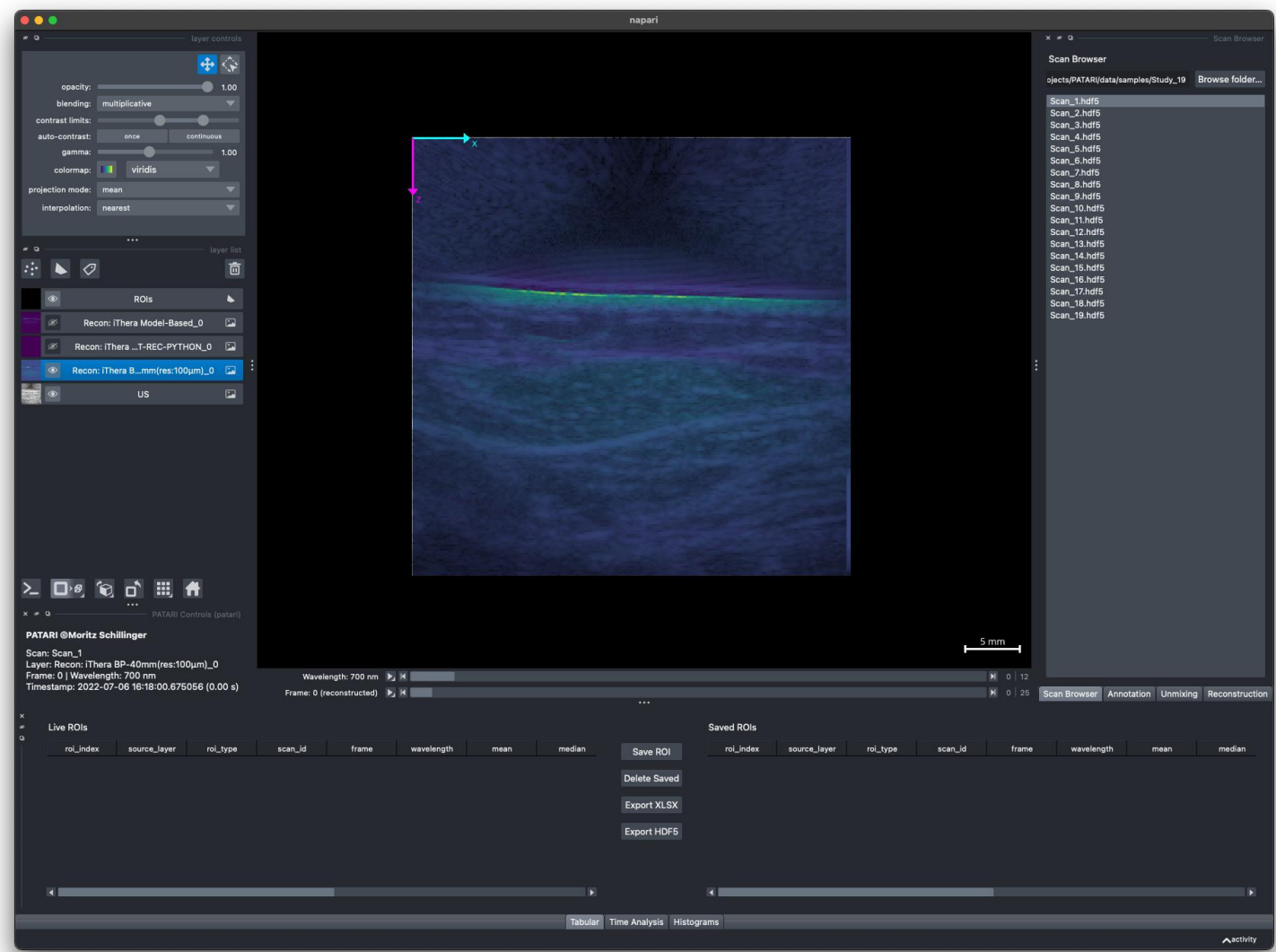
Screenshots



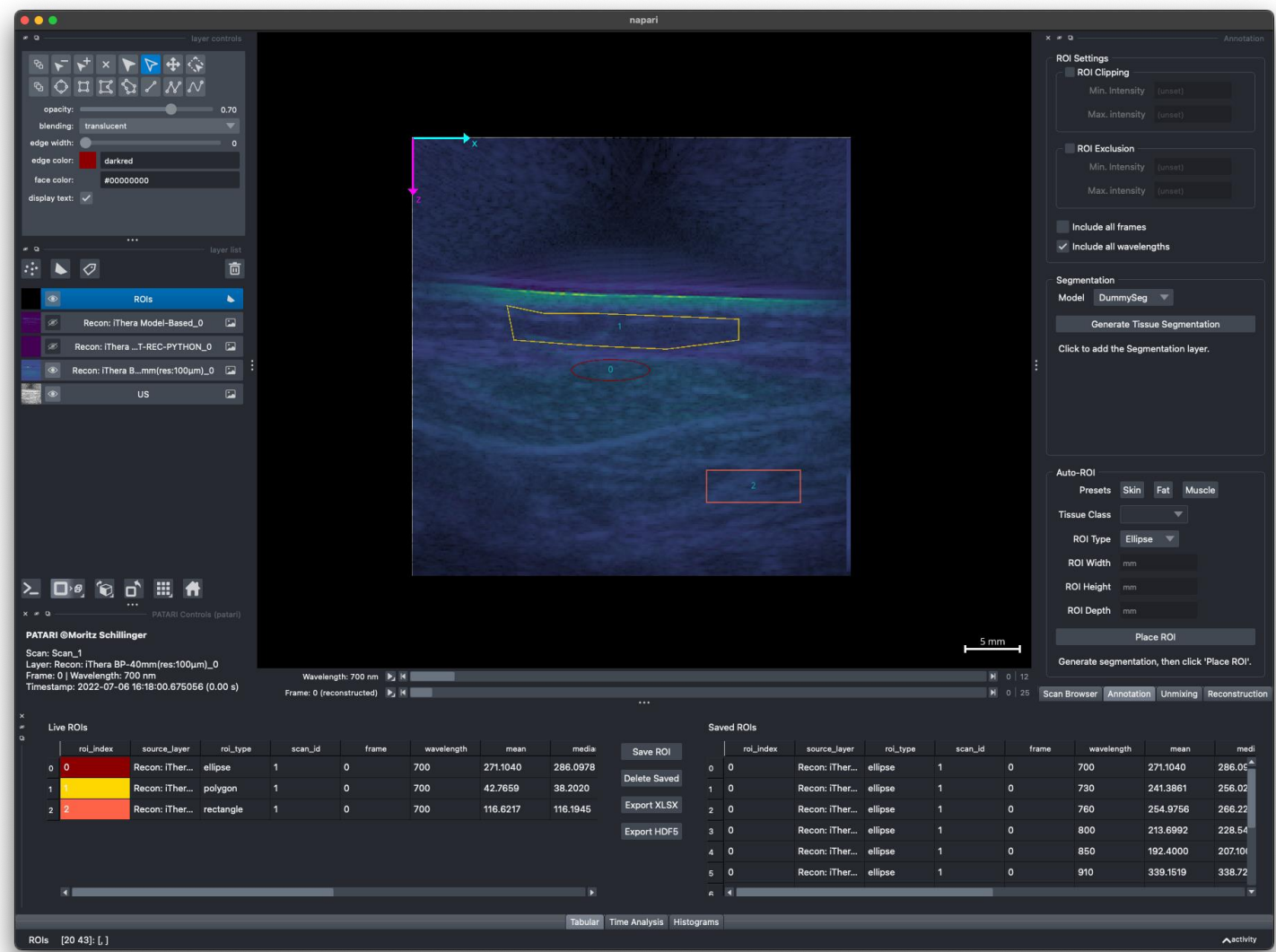
Side by Side View + Scan Browser



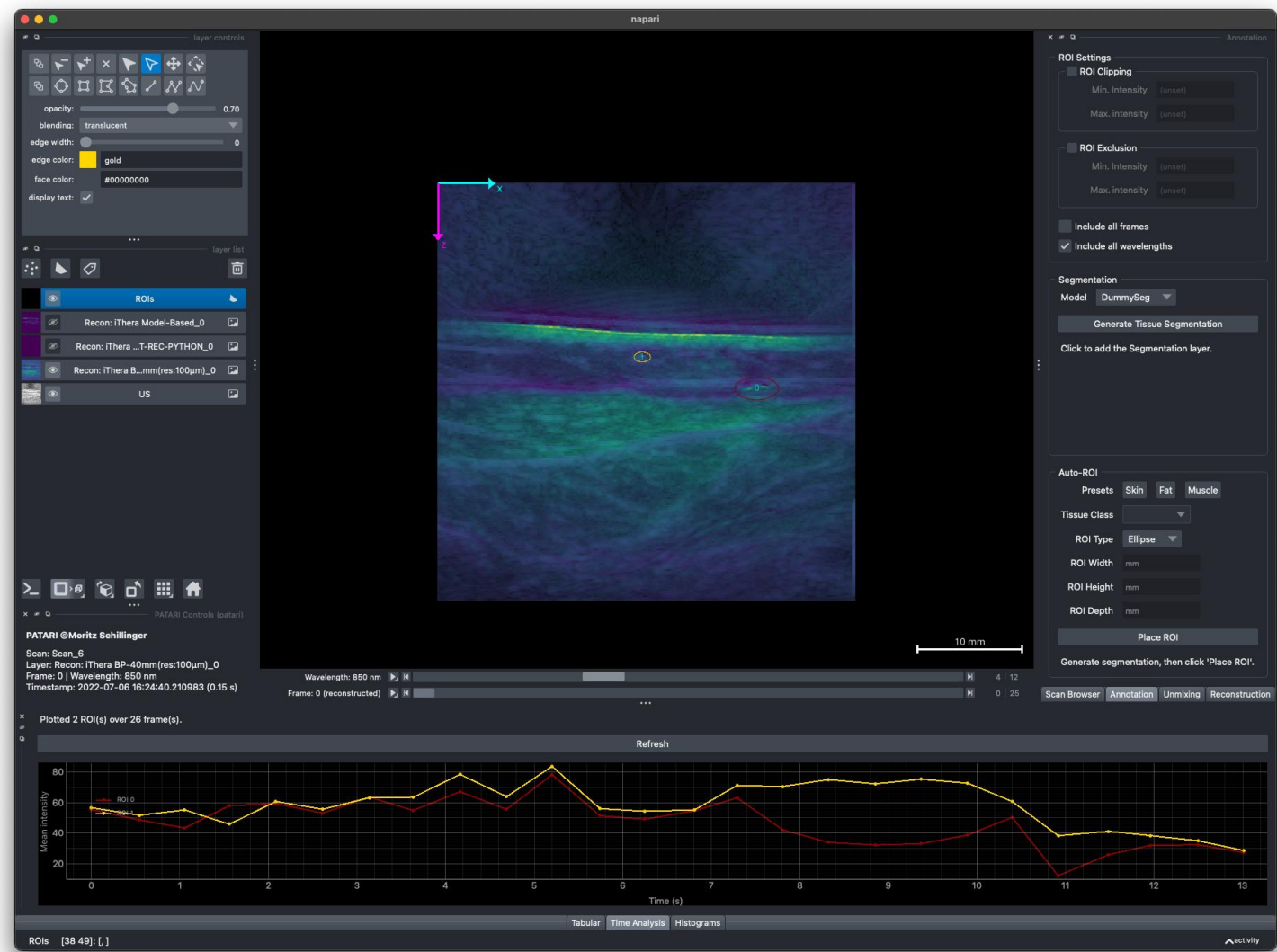
Layer Overlay



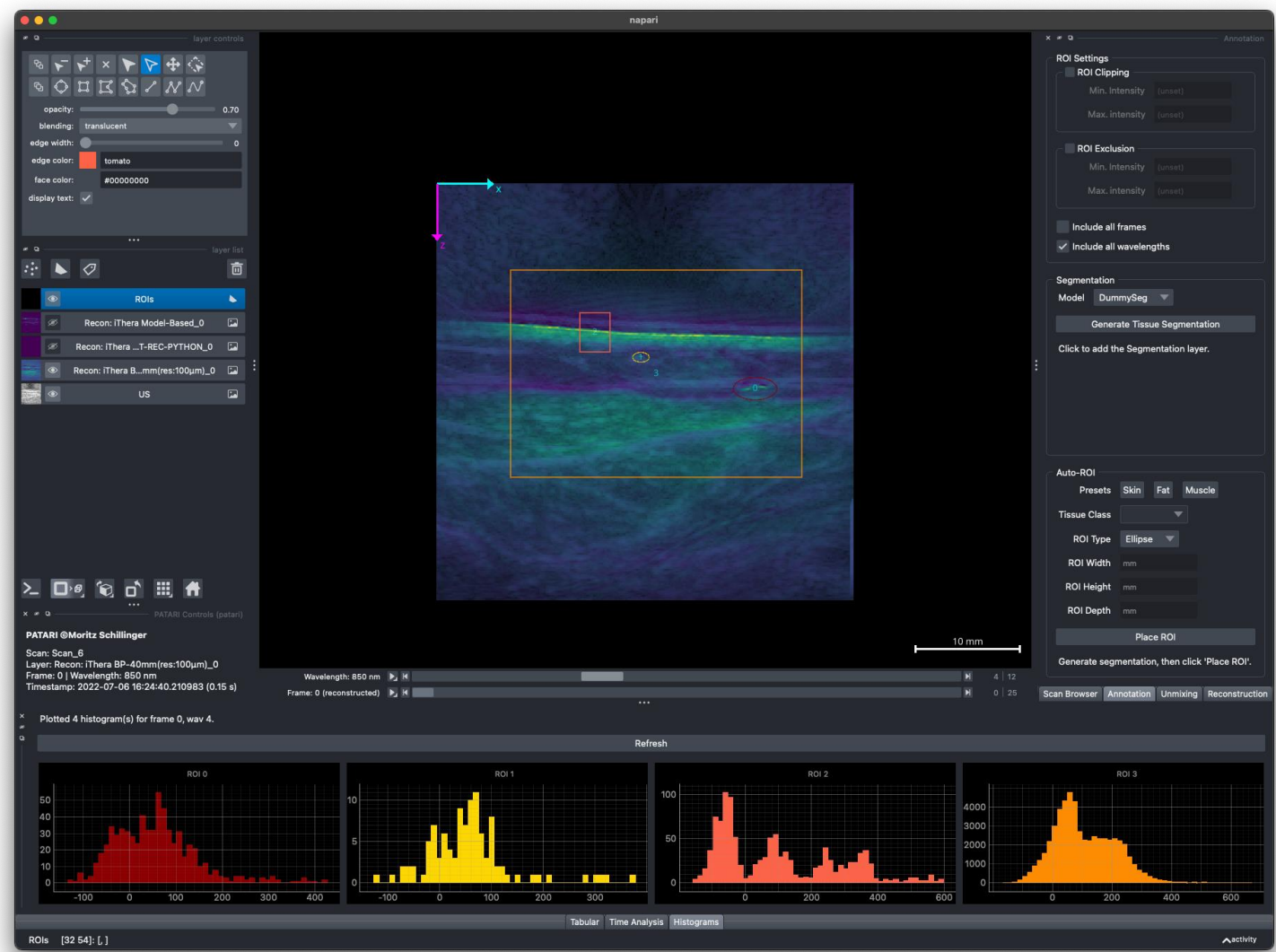
ROI Placement + Intensity Extraction + Live / Saved Table

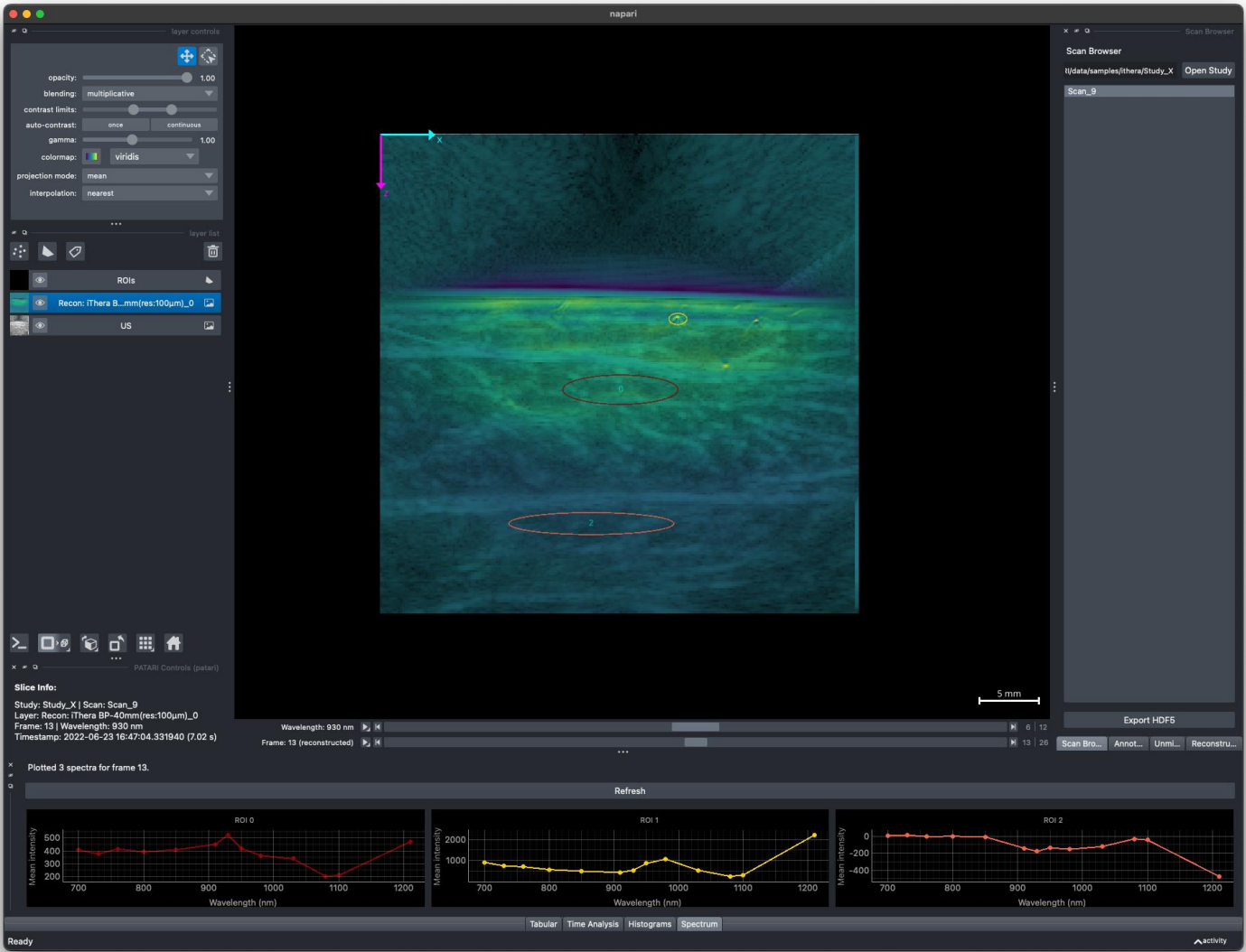


Intensity in ROI over Time



Histogram View





DL-based Tissue Segmentation + Automatic ROI

